

Topic 5: Arms Race

Zeyn Jaswal, Azzaam Khan, Timothy Palin, Meredith Reeves, and
Tony Xu

McMaster University

March 26, 2026



Overview



Scenario: Purple (P) and Green (G) are two countries competing in an arms race.

Research Question: What happens to the long-term spending of each country?

Key Factors:

- More inclined to increase defence spending as the other's rises
- Less inclined to increase spending as its own spending rises
- Does public sentiment affect defence spending?
- Does limiting spending affect a country's position?

Motivation: It helps predict military escalation between countries, which can inform policies to promote peace and prevent conflict.

Base Model



$$\frac{\partial p}{\partial t} = ag(t) - bp(t)$$
$$\frac{\partial g}{\partial t} = cp(t) - dg(t)$$

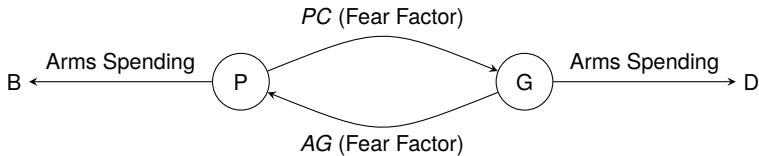


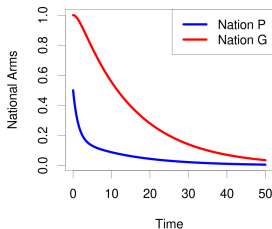
Figure: Base Model

Long Term Behavior (Base Model)



$$bd > ac$$

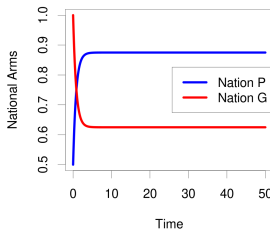
$$a=0.1, b=0.7, c=0.2, d=0.1$$



Complete mutual
disarmament

$$bd = ac$$

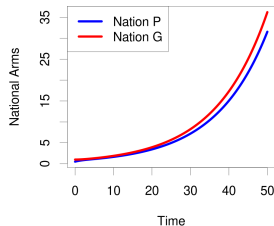
$$a=0.7, b=0.5, c=0.5, d=0.7$$



Stabilized spending

$$bd < ac$$

$$a=0.5, b=0.5, c=0.2, d=0.1$$



Perpetual arms race

Public Sentiment (Trust and Hostility)



$$\frac{\partial p}{\partial t} = ag(t) - bp(t) + r$$
$$\frac{\partial g}{\partial t} = cp(t) - dg(t) + s$$

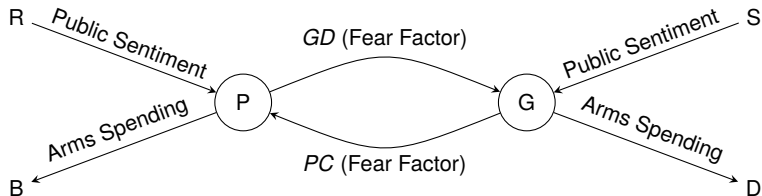


Figure: Base Model Including Public Sentiment

Long Term Behavior (Public Sentiment)



- provide plots
- provide stability conditions and convergences
- This model will take on negative values so discuss that by either capping the value or other methods

Mutual Logistic Fear

Limitation: Add logistic growth to account for limited resources.

$$\frac{\partial p}{\partial t} = a(1 - \frac{p(t)}{K_p})g(t) - bp(t) + r$$

$$\frac{\partial g}{\partial t} = c(1 - \frac{g(t)}{K_g})p(t) - dg(t) + s$$

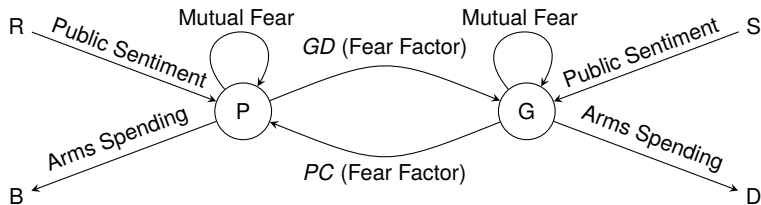
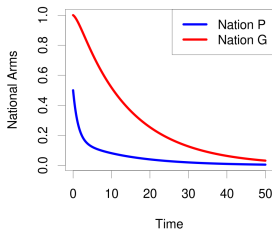


Figure: Base Model Including Public Sentiment and Mutual Fear

Long Term Behavior (Mutual Logistic Fear)

$$bd > ac$$

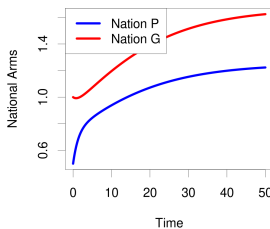
$$a=0.1, b=0.7, c=0.2, d=0.1, K_p=5, K_g=5$$



Complete mutual
disarmament

$$bd = ac$$

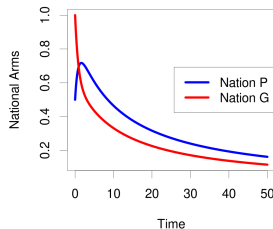
$$a=0.5, b=0.5, c=0.2, d=0.1, K_p=5, K_g=5$$



Stabilized spending

$$bd < ac$$

$$a=0.7, b=0.5, c=0.5, d=0.7, K_p=5, K_g=5$$



Perpetual arms race

Thank you for listening!



[©] Presentation theme by Ricardo B. Lourenco